DAXALGO RESEARCH // REVISION REPORT

CHMT-Q v1 failed as a trading model

Deep failure analysis and revision design | OKX BTC-USDT | June 2026

DECISION

DO NOT DEPLOY. DO NOT ALLOCATE CAPITAL.

The checkpoint is intact; the research design is misaligned with executable return and underpowered.

+2.26%	+0.01%	0	-68.1%
aggregate pinball	**H5 POC median MAE**	**primary-policy trades**	**base-cost return**
vs published marginal baseline	effectively tied with baseline	all intervals crossed zero	median long/flat proxy

WHAT FAILED

Objective

Five footprint distributions, not return, price, action, or net PnL.

Signal

H5 close-direction MCC 0.0475; POC Spearman 0.078.

Economics

0.45 bps gross round-trip edge versus 24 bps modeled cost.

Evidence

18 train days, 2 selection days, 4 forward days, one seed and fold.

Revision status: diagnosis complete; replacement model is not yet trained or validated.

01 // ROOT-CAUSE MAP

The failure is a chain, not one bad hyperparameter

Every link is observed in the authorized Jun 1-24 evidence; causal strength is labeled explicitly.

1 WRONG ENDPOINT

Footprint pinball

No direct return target

→

2 WEAK MEDIAN

H5 POC rho 0.075

Std ratio 0.151

→

3 WIDE RISK

$190.47 interval

100% crosses zero

→

4 ABSTENTION

0 primary trades

Correct under uncertainty

→

5 COST FAILURE

0.45 vs 24 bps

-68.1% base cost

Ranked evidence

PROVEN	Objective does not optimize executable return.
PROVEN	Gross edge is 53.9x below the one-way base-cost hurdle.
PROVEN	Three POC heads are redundant and presence is constant.
PROVEN	Every update was gradient-clipped; schedules stopped at 36%.
PROVEN	Sparse L1 changes lose magnitude at scaler boundaries.
LIKELY	Selection optimism, regime shift, and overfit coexist.

What did not fail

- Checkpoint and source hashes match.
- Scaler is bound to training IDs.
- Horizon alignment reconstructs exactly.
- No NaN or attention-logit runaway.
- No evidence of a compiler off-by-one.

Conclusion: the artifact is internally consistent, but consistency cannot repair a target, sample, and execution design that does not identify tradable edge.

The headline score hides the target that matters

Positive means model improvement over the published empirical marginal; negative means degradation.

3.46%

objective weight

H5 total-POC quantile cell

+2.26%

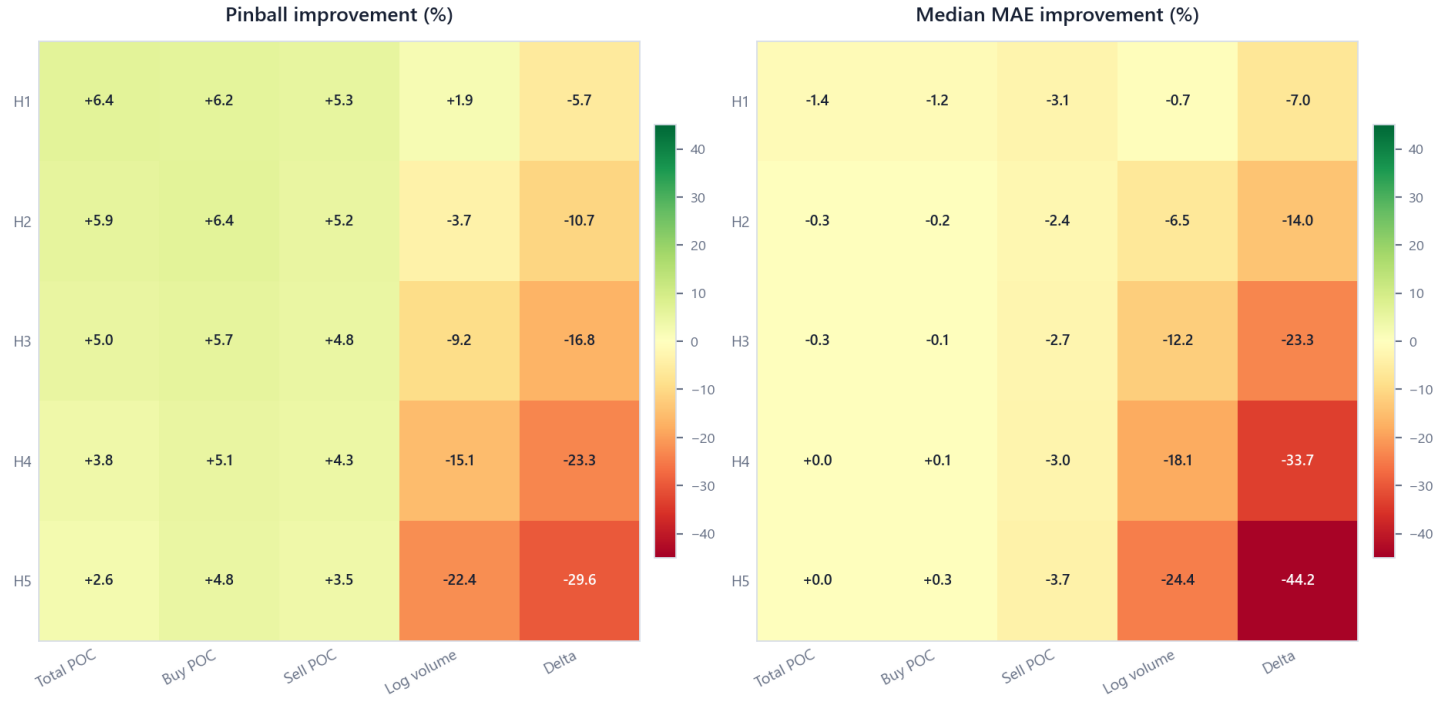
aggregate pinball

all horizons and targets

75%

forecast weight

three highly correlated POC heads

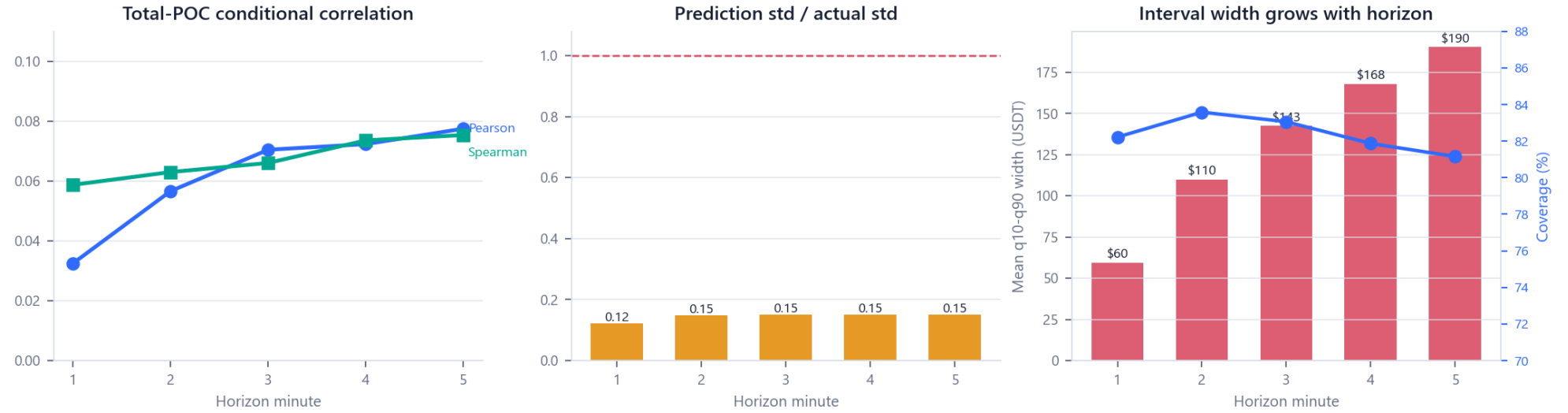


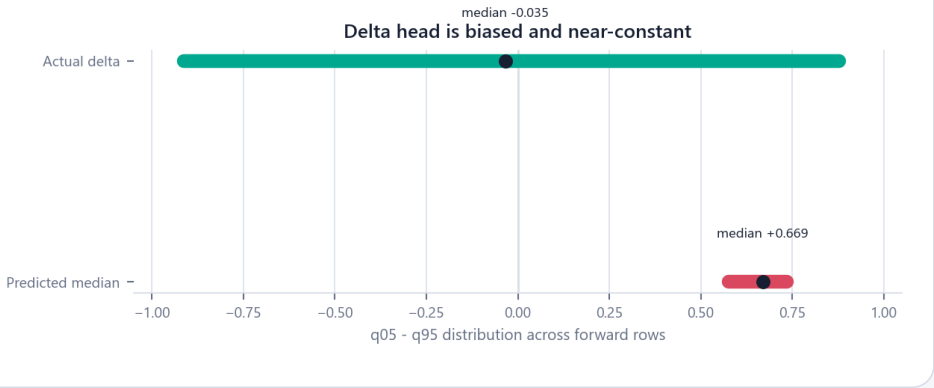
H5 result: total POC pinball +2.62% and median MAE +0.01%; log-volume pinball -22.43% / MAE -24.44%; delta pinball -29.57% / MAE -44.18%.

Model parameters: 2,089,554 | target count: 5 x 5 x 3 quantiles

Forecasts are smooth, weak, and too uncertain to trade

Forward period: Jun 21-24 only. POC offsets are converted from ticks using the 0.1 USDT tick size.





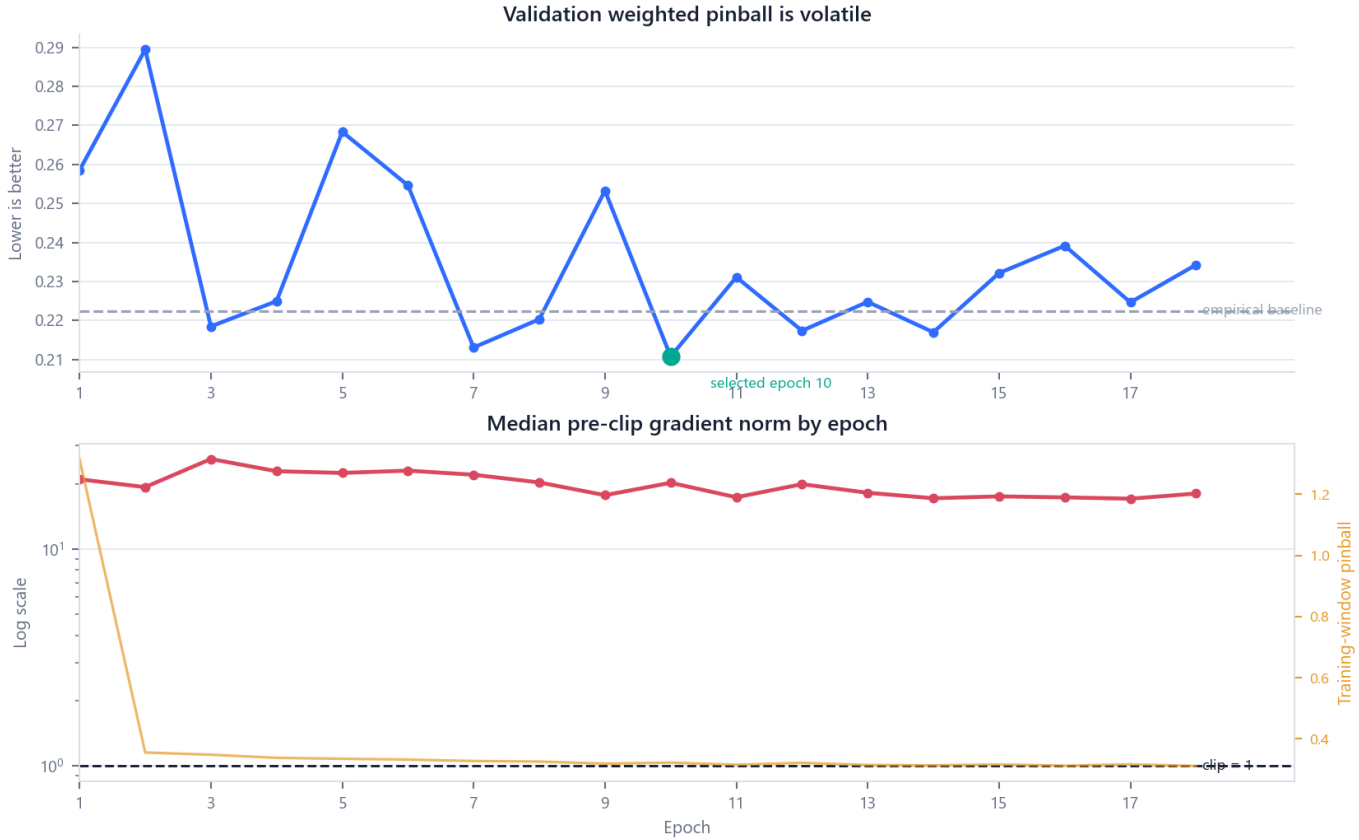
Five-minute diagnostics

POC Pearson / Spearman	**0.077 / 0.075**
Prediction std / actual std	**0.151**
Mean q10-q90 width	**190.47 USDT**
Intervals crossing zero	**100.0%**
Delta median bias / correlation	**+0.688 / -0.011**

Calibration coverage alone is not useful when every actionable interval includes zero.

Training selected a volatile checkpoint before schedules matured

Mechanics are proven; their causal effect requires controlled ablation.



100%

updates clipped

median norm 19.58 vs clip 1.0

36%

schedule completed

3,618 of 10,050 planned steps

0.25

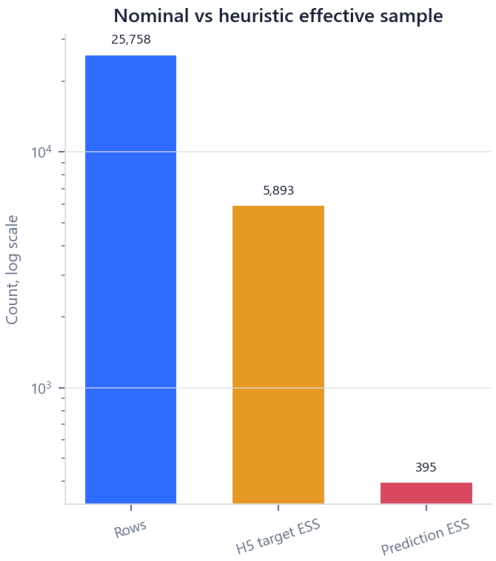
auxiliary weight

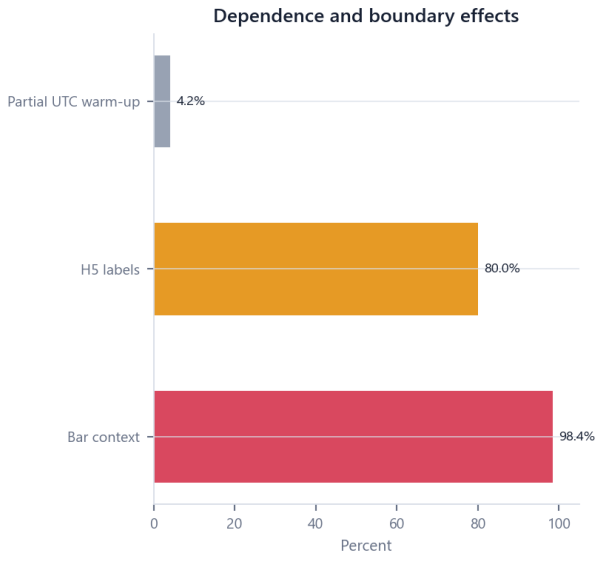
never decayed toward 0.05

Best epoch 10: 0.210765. Final epoch 18: 0.234276 (11.2% worse). Training-window pinball improved 0.321693 -> 0.310601. Final LR 2.35e-04; intended floor 3.0e-05.

25,758 rows are not 25,758 independent markets

Dependence, scale collapse, and narrow regimes reduce the information available to a 2.09M-parameter model.





Pipeline defects and validity limits

- 98.4% adjacent bar-context overlap; 80% H5-label overlap.
- Five future-minute reuses: 171,720 occurrences -> 34,440 unique minutes.
- L1 mid/bid/ask changes clip at about 22.84% each.
- UTC reset leaves 4.19% of rows with partial history.
- No independent receive time or native L2 sequence/checksum evidence.
- Event time substitutes for availability; decision lag is 0 ms.

18

training sessions

plus 2 selection / 4 forward

81.1

parameters per row

116,086 per session

1 seed

one fold trained

no variance estimate

4 days

forward evidence

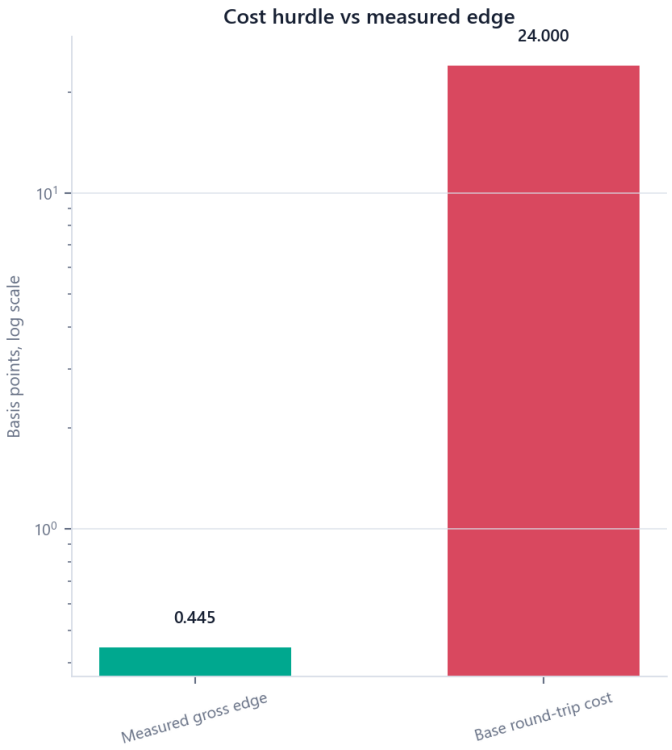
insufficient for regime/tail claims

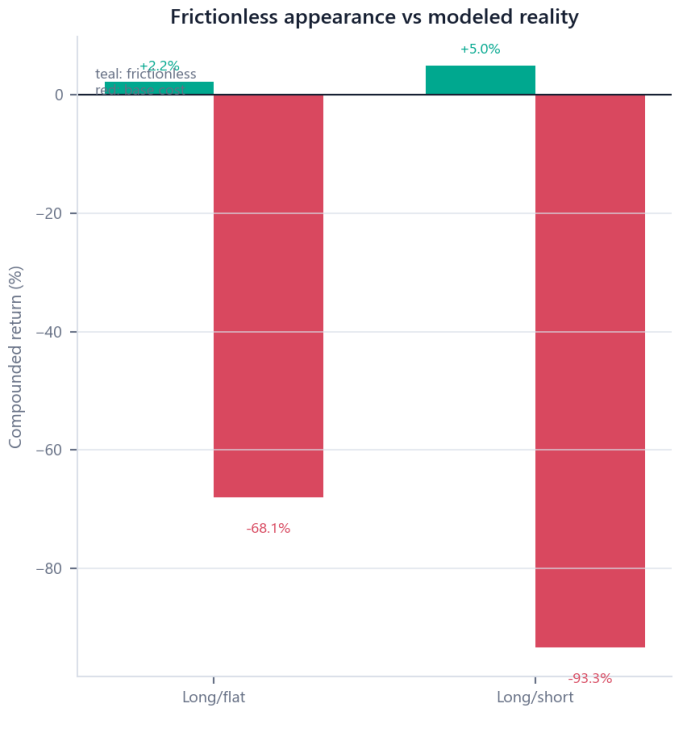
Holdout integrity disclosure: Jun 25-30 target fields were inadvertently read during an auxiliary schema/count audit, not retained or used. Treat that period as process-accessed and replace it for final confirmation.

06 // ECONOMIC FAILURE

Transaction costs are not a sensitivity; they are the result

Five-minute non-overlapping execution proxy. Base cost is 12 bps per side, 24 bps round trip.





53.9x

cost / break-even

median long/flat

0.0475

direction MCC

accuracy 52.67%

0

primary trades

uncertainty-aware interval rule

Long/flat: 484 trades, frictionless +2.16%, base -68.07%, break-even 0.223 bps/side. Long/short: 1148 trades, frictionless +4.99%, base -93.34%.

Revision: predict executable net return, keep footprint as auxiliary

Start with strong simple baselines; retain a transformer only if paired evidence justifies it.

QUALIFIED DATA

3-6+ months
receive time + seq/checksum
executable bid/ask

>

FEATURE LAYER

tick-aware sparse scaling
5-minute decisions
continuous warm-up

>

MODEL LADDER

linear / trees / TCN first
then 0.2-0.5M encoder
modality ablations

>

RETURN HEAD

long/short net-return
q10 / q50 / q90
volatility + downside

—

AUX HEAD

POC / volume / delta
low weight only
no presence task

Executable target and decision rule

Long target: log(exit bid / entry ask) - fee - slippage - financing.

Short target: log(entry bid / exit ask) - fee - slippage - financing.

Action: long only if q10(long net return) > 0; short only if q10(short net return) > 0; otherwise flat.

Size: capped by predicted volatility, interval width, liquidity, and portfolio risk budget.

Preprocessing fixes

Separate zero/change indicator from signed magnitude; use tick-aware or quantile transforms; require <1% feature clipping; carry 64-minute warm-up across UTC boundaries; use strict < cutoff and realistic decision latency.

Optimization fixes

Re-budget learning-rate and auxiliary schedules for the real stopping horizon; normalize losses by target; remove constant labels; target <10% post-warmup clipping; compare auxiliary-on/off and every modality with paired folds and seeds.

A gated experiment, not another single training run

Each phase has an explicit stop condition. Thresholds below are proposed and must be frozen before final confirmation.

A DATA
quality + causality
new sealed period

B BASELINES
recent/hour/ridge/tree
best simple model

C MODELS
TCN + compact encoder
3 folds x 3 seeds

D CALIBRATE
interval + threshold
frozen policy

E CONFIRM
new holdout once
no retuning

F SHADOW
live paper execution
no capital

Go / no-go table

GATE	PASS CONDITION	EVIDENCE
Forecast	>=5% proper-score gain vs strongest simple baseline	every fold; seeds 17/29/43
Direction	MCC >0.10 with stable sign	not accuracy alone
Economics	gross edge >=3x frozen all-in cost	base and stress costs
Uncertainty	80% interval coverage in 76-84% band	competitive width + active trades
Statistics	day-block lower CI >0 for net return	>=60 untouched days
Optimization	post-warmup clip <10%; feature clip <1%	schedules reach intended phase
Process	zero causal violations; new holdout opened once	no threshold retuning

FAIL ANY GATE -> STOP. DO NOT OPEN THE NEW HOLDOUT.

Audit conclusion and handoff

What is proven, what remains hypothetical, and what must happen next.

Proven from this run

- Objective is footprint prediction, not executable return.
- Median signal is weak; uncertainty crosses zero.
- Costs exceed measured edge by more than 50x.
- Targets are redundant/degenerate; sparse scaling loses magnitude.
- Training is clipping-dominated and stopped before schedule decay.

Not yet proven

- That any revised model will be profitable.
- That the transformer beats strong simple baselines.
- That L2 or the hierarchy adds incremental value.
- That clipping/scaling fixes alone restore performance.
- That simulated fills transfer to live execution.

Holdout integrity

An early schema/count projection traversed all 42,930 Arrow rows and physically read projected target fields in the Jun 25-30 shards. No sealed values were printed, retained, analyzed, or used. The period must nevertheless be treated as process-accessed and replaced for future confirmatory testing.

Action: retire Jun 25-30 as a pristine confirmation set and seal a new later period before training begins.

Historical backtest outer_holdout_accessed=false remains true for artifact creation; it is not a current-session integrity claim.

Artifact lineage

DATASET	565c1d8e912ee426a1a92572d5be481ec5168d54d1ae53bd77fdd70d888e4df6
EXPERIMENT	25be2f446a75ad37120bba956ea86e46c0ecefa004ab3343d6745cdbee102fb3
CHECKPOINT	step-00002010
NUMERIC SOURCE	7bd3153371f1e09d0aa33d924c5487111efd36589b6b4a498fe78175a48670d3

FINAL DECISION: RESEARCH REDESIGN ONLY

NO DEPLOYMENT / NO CAPITAL